

## FORT SEVERN, ON, Canada

WMO: 710990

Lat: 56.020N

Lon: 87.680W

Elev: 16

StdP: 101.14

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD<br>to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|--------------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                          |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                     | PCWD       |              |
| (a)<br>1      | (b)<br>-36.7 | (c)<br>-34.4 | (d)<br>-40.9                  | (e)<br>0.1 | (f)<br>-36.7 | (g)<br>-38.6 | (h)<br>0.1 | (i)<br>-34.3 | (j)<br>10.4           | (k)<br>-19.4 | (l)<br>9.8 | (m)<br>-20.4 | (n)<br>3.2               | (o)<br>250 | (p)<br>0.867 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>11.6            | (c)<br>26.9     | (d)<br>18.3 | (e)<br>24.2 | (f)<br>17.0 | (g)<br>21.7 | (h)<br>15.9 | (i)<br>19.4         | (j)<br>24.5 | (k)<br>17.8 | (l)<br>22.8 | (m)<br>16.5 | (n)<br>20.8 | (o)<br>5.9           | (p)<br>240 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.2                            | 12.3 | 22.0 | 15.8 | 11.2 | 20.1 | 14.4 | 10.3 | 18.7 | 55.7          | 24.3 | 50.3 | 22.9 | 46.5 | 20.7 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 10.3          | (b) 9.2 | (c) 8.3 | (e) -40.3                  | (f) 31.5 | (g) 3.1            | (h) 2.5 | (i) -42.6                                          | (j) 33.3 | (k) -44.4  | (l) 34.8 | (m) -46.2  | (n) 36.2 | (o) -48.4  | (p) 38.0 |    |    |
| (a) 10.3          | (b) 9.2 | (c) 8.3 | (e) -40.8                  | (f) 22.0 | (g) 3.2            | (h) 1.6 | (i) -43.1                                          | (j) 23.1 | (k) -45.0  | (l) 24.0 | (m) -46.8  | (n) 24.9 | (o) -49.1  | (p) 26.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |                                                                                  | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |                                                                                  | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg                                                                            | -4.1   | -23.1 | -22.4 | -15.8 | -7.5  | 1.3   | 7.9   | 12.7  | 12.9  | 8.5   | 2.0   | -8.2  | -18.4 |      |
|                                                                                  | DBStd                                                                            | 14.24  | 6.95  | 6.05  | 7.73  | 6.82  | 5.00  | 5.10  | 4.38  | 3.89  | 4.04  | 3.92  | 6.92  | 8.63  |      |
|                                                                                  | HDD10.0                                                                          | 5418   | 1027  | 908   | 799   | 524   | 275   | 102   | 19    | 12    | 77    | 251   | 546   | 879   |      |
|                                                                                  | HDD18.3                                                                          | 8197   | 1286  | 1141  | 1058  | 774   | 527   | 317   | 183   | 177   | 296   | 507   | 796   | 1138  |      |
|                                                                                  | CDD10.0                                                                          | 284    | 0     | 0     | 0     | 0     | 6     | 39    | 104   | 102   | 32    | 2     | 0     | 0     |      |
|                                                                                  | CDD18.3                                                                          | 22     | 0     | 0     | 0     | 0     | 0     | 4     | 9     | 8     | 1     | 0     | 0     | 0     |      |
|                                                                                  | CDH23.3                                                                          | 313    | 0     | 0     | 0     | 0     | 4     | 77    | 142   | 82    | 8     | 1     | 0     | 0     |      |
|                                                                                  | CDH26.7                                                                          | 79     | 0     | 0     | 0     | 0     | 0     | 21    | 39    | 18    | 1     | 0     | 0     | 0     |      |
| Wind                                                                             | WSAvg                                                                            | 4.3    | 4.0   | 4.0   | 4.3   | 4.4   | 4.5   | 4.5   | 4.0   | 4.2   | 4.4   | 4.5   | 4.5   | 4.2   |      |
| Precipitation                                                                    | PrecAvg                                                                          | 549    | 19    | 16    | 21    | 24    | 46    | 61    | 81    | 80    | 73    | 62    | 40    | 25    |      |
|                                                                                  | PrecMax                                                                          | 665    | 37    | 40    | 43    | 59    | 99    | 106   | 159   | 179   | 171   | 147   | 75    | 68    |      |
|                                                                                  | PrecMin                                                                          | 365    | 5     | 3     | 1     | 8     | 14    | 24    | 33    | 38    | 27    | 19    | 18    | 6     |      |
|                                                                                  | PrecStd                                                                          | 77     | 8     | 9     | 10    | 12    | 22    | 21    | 31    | 29    | 32    | 28    | 17    | 15    |      |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                             | DB     | -1.3  | -2.8  | 7.1   | 10.6  | 23.3  | 29.3  | 30.4  | 29.1  | 24.4  | 16.8  | 6.7   | -0.2  |      |
|                                                                                  |                                                                                  | MCWB   | -1.9  | -3.7  | 4.0   | 5.7   | 13.7  | 18.1  | 20.1  | 20.9  | 17.4  | 12.3  | 4.3   | -0.7  |      |
|                                                                                  | 2%                                                                               | DB     | -5.0  | -6.7  | 3.8   | 7.6   | 17.6  | 25.6  | 27.3  | 25.5  | 21.7  | 13.0  | 3.3   | -1.9  |      |
|                                                                                  |                                                                                  | MCWB   | -5.3  | -7.7  | 1.4   | 3.5   | 10.9  | 16.4  | 18.3  | 18.0  | 16.2  | 10.0  | 2.1   | -2.2  |      |
|                                                                                  | 5%                                                                               | DB     | -8.1  | -10.0 | 0.4   | 5.2   | 12.7  | 21.4  | 24.2  | 22.9  | 18.1  | 9.6   | 1.5   | -3.5  |      |
|                                                                                  |                                                                                  | MCWB   | -8.4  | -10.8 | -1.2  | 2.1   | 8.1   | 14.6  | 17.4  | 17.2  | 14.0  | 7.9   | 0.7   | -3.9  |      |
|                                                                                  | 10%                                                                              | DB     | -12.0 | -13.0 | -3.6  | 2.6   | 8.7   | 17.6  | 21.4  | 19.9  | 15.3  | 7.0   | 0.3   | -5.6  |      |
|                                                                                  |                                                                                  | MCWB   | -12.5 | -13.7 | -4.8  | 0.5   | 5.9   | 13.2  | 16.1  | 15.9  | 12.1  | 5.7   | -0.3  | -6.0  |      |
|                                                                                  | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | -1.7  | -3.6  | 4.0   | 6.3   | 13.8  | 20.5  | 21.6  | 21.2  | 18.5  | 13.5  | 4.7   | -0.3 |
|                                                                                  |                                                                                  |        | MCDB  | -1.3  | -3.0  | 6.6   | 10.3  | 22.1  | 26.2  | 27.3  | 27.8  | 22.7  | 16.8  | 6.1   | 0.2  |
| 2%                                                                               |                                                                                  | WB     | -5.3  | -7.6  | 1.5   | 3.7   | 11.0  | 17.2  | 19.5  | 19.1  | 16.2  | 10.4  | 2.4   | -2.0  |      |
|                                                                                  |                                                                                  | MCDB   | -5.1  | -6.8  | 3.5   | 6.9   | 17.3  | 23.6  | 25.1  | 23.5  | 20.6  | 13.0  | 3.3   | -1.8  |      |
| 5%                                                                               |                                                                                  | WB     | -8.5  | -10.8 | -1.3  | 2.4   | 8.4   | 15.3  | 17.8  | 17.6  | 14.2  | 7.9   | 0.9   | -3.6  |      |
|                                                                                  |                                                                                  | MCDB   | -8.2  | -10.0 | 0.5   | 5.0   | 12.2  | 20.9  | 23.1  | 22.2  | 17.9  | 9.9   | 1.5   | -3.3  |      |
| 10%                                                                              |                                                                                  | WB     | -12.4 | -13.7 | -4.7  | 0.5   | 6.1   | 12.8  | 16.5  | 16.2  | 12.3  | 5.8   | -0.1  | -5.7  |      |
|                                                                                  |                                                                                  | MCDB   | -12.0 | -13.0 | -3.5  | 2.6   | 9.0   | 17.9  | 20.7  | 19.7  | 15.0  | 6.9   | 0.4   | -5.3  |      |
| Mean Daily<br>Temperature<br>Range                                               |                                                                                  | 5% DB  | MDBR  | 8.9   | 10.3  | 12.0  | 10.0  | 8.4   | 10.8  | 11.6  | 9.5   | 8.3   | 5.5   | 7.1   | 7.6  |
|                                                                                  |                                                                                  |        | MCDBR | 10.7  | 12.8  | 13.6  | 12.7  | 16.8  | 17.7  | 16.9  | 14.4  | 13.6  | 9.8   | 6.3   | 6.3  |
|                                                                                  | MCWBR                                                                            |        | 10.7  | 12.0  | 11.7  | 9.3   | 10.8  | 9.8   | 9.6   | 8.1   | 8.7   | 7.3   | 5.5   | 6.3   |      |
|                                                                                  | 5% WB                                                                            | MCDBR  | 10.7  | 12.9  | 13.1  | 11.9  | 16.4  | 16.8  | 15.2  | 12.9  | 12.7  | 9.3   | 6.0   | 6.1   |      |
|                                                                                  |                                                                                  | MCWBR  | 10.7  | 12.2  | 11.3  | 9.0   | 10.6  | 10.4  | 9.5   | 7.8   | 8.7   | 7.2   | 5.4   | 6.1   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub                                                                             |        | 0.234 | 0.257 | 0.282 | 0.360 | 0.382 | 0.372 | 0.401 | 0.373 | 0.339 | 0.302 | 0.259 | 0.216 |      |
|                                                                                  | taud                                                                             |        | 2.134 | 2.173 | 2.167 | 2.005 | 2.109 | 2.274 | 2.316 | 2.418 | 2.501 | 2.522 | 2.288 | 2.126 |      |
|                                                                                  | Ebn at Noon                                                                      |        | 683   | 816   | 883   | 848   | 853   | 869   | 833   | 838   | 823   | 775   | 656   | 614   |      |
|                                                                                  | Edh at Noon                                                                      |        | 57    | 82    | 108   | 149   | 144   | 124   | 117   | 99    | 80    | 61    | 49    | 44    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg                                                                           |        | 0.83  | 1.91  | 3.50  | 5.10  | 5.44  | 5.96  | 5.39  | 4.12  | 2.66  | 1.38  | 0.61  | 0.51  |      |
|                                                                                  | RadStd                                                                           |        | 0.07  | 0.14  | 0.23  | 0.41  | 0.49  | 0.57  | 0.48  | 0.37  | 0.28  | 0.12  | 0.10  | 0.07  |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | -41     | N/A     |

Nomenclature: See separate page